

AgroStruxure™

Solar-Synchronized Agri-Energy Microgrid & Precision Irrigation

"One solar pump can become a rural energy node."

Decoupling agricultural water pumping from free solar energy to eliminate the **Solar Rebound Paradox** across 10.9 lakh PM-KUSUM installations. AgroStruxure dynamically diverts idle midday solar generation into farm-gate thermal pre-cooling while automating closed-loop groundwater conservation.

41.2%

WATER CONSERVED

7,000 m³/ha/yr preserved via dual-depth soil cutoff

4,137 kWh

SOLAR SURPLUS CAPTURED

63% wasted PV redirected to 2 MT pre-cooling

2.0 Yrs

CLUSTER PAYBACK

Tier A 4-farm asset sharing at ₹50,212/farm capex



Core Engineering Team: Sanskar Tiwari (Firmware/Lead) • Shambhavi Patil (Geospatial) • Kanishka Salgude (Power/VFD) • Chaitanya Ranade (Agronomy/Product)



FIELD ASSET ARCHITECTURE

PM-KUSUM Component B: 4.8 kWp Standalone PV

Ground-mounted perimeter array powering 3-phase submersible pump with automated upstream DC changeover to 2 MT thermal PCM cold storage.

PV Array: **4.8 kWp (DC)**

Annual Yield: **6,559 kWh**

Bus Voltage: **350-600V DC**

Telemetry: **Modbus RTU / 4G**

Free Solar Power Accelerates Groundwater Collapse

Under PM-KUSUM Component B, zero marginal electricity cost removes the economic friction on pumping, triggering a triple cascading bottleneck across water reserves, clean energy utility, and smallholder income.



01

Unconstrained Extraction & Root Waterlogging

Farmers operate pumps whenever sunlight is available (850 mm/season across flood basins). This floods root zones, inducing hypoxia, fungal collar rot, nutrient leaching, and massive aquifer depletion without yielding additional crop tonnage.

17,000
M³/HA/YR EXTRACTED

↓ ZERO PUMPING FRICTION SATURATES FIELD STORAGE WITHOUT COLD CHAIN ↓

02

4,137 kWh of Solar Energy Wasted Idle Annually

A 4.8 kWp standalone array generates 6,559 kWh/year. Pumping only requires 2,422 kWh/year. Corroborating IWMI findings (Shah et al.), **nearly two-thirds of capital-subsidized solar power sits unused** once storage ditches reach capacity.

63.1%
SOLAR ENERGY
CURTAILMENT

↓ UNSTABLE FIELD HEAT ACCELERATES POST-HARVEST RESPIRATION ↓

03

8.37% Farm-Gate Spoilage from Missing Cold Chain

Per NABCONS 2022 Post-Harvest Loss Study, 8.37% of harvested tomatoes rot before leaving the farm gate due to extreme 32°C respiration heat. Farmers face severe distress selling at local mandis (₹2–₹4/kg vs ₹12/kg market benchmark).

8.37%
PERISHABLE CROP LOST

Industrial-grade retrofit marrying ESP32-S3 edge intelligence with Schneider Altivar Solar VFDs and TeSys switchgear, operating autonomously without cloud dependency.

41.2% Less Groundwater. Zero Compromise on Yield.

Combining FAO-56 dual Kc dynamic evapotranspiration budgeting with real-time FDR capacitive cutoff eliminates agricultural over-irrigation.

GROUND TRUTH PHYSICS

Dual-Depth Soil Moisture Profile

Continuous capacitive frequency domain reflectometry (FDR) prevents irrigation beyond Field Capacity (FC), eliminating deep percolation losses.

0–10 cm: Surface Evaporative Zone

Tracked by 10cm FDR Probe • High diurnal vapor loss

18%–32% VWC

10–40 cm: Active Crop Root Zone

Dynamic Cutoff Triggered at FC = 45% • Zero Hypoxia

FC CUTOFF

40+ cm: Deep Percolation Saturated Zone

Eliminated by pulsed drip • Previously 7,000 m³ drained uselessly

NO LEACHING



Hydraulic Solenoid Interlock: Closes automatically upon root-zone field capacity saturation. Prevents collar rot in Solanaceae crops and halts unnecessary pump power draw.

IMPACT_MODEL.PY QUANTITATIVE PROOF

Water & Energy Conservation Ledger (Per Ha / Year)

PARAMETER	BASELINE FLOOD	AGROSTRUXURE	NET SAVINGS
Water Application	17,000 m ³	10,000 m ³	-7,000 m ³ (-41.2%)
Pumping Electricity	4,118 kWh	2,422 kWh	-1,696 kWh Liberated
Pumping Operating Time	1,373 hrs	807 hrs	-566 hrs Less Wear
Root Zone Health	Hypoxia / Leaching	Aerobic / Optimal	Zero Waterlogging

CONSERVED GROUNDWATER PER HECTARE

7,000 m³ / Year

Equivalent to 28 Olympic swimming pools of community aquifer water preserved.

41.2%

DIRECT REDUCTION

Honest Attribution: 29.4% saved by shifting flood to drip; 11.8% additional saving uniquely locked in by AgroStruxure's automated soil cutoff to eliminate the solar rebound paradox.

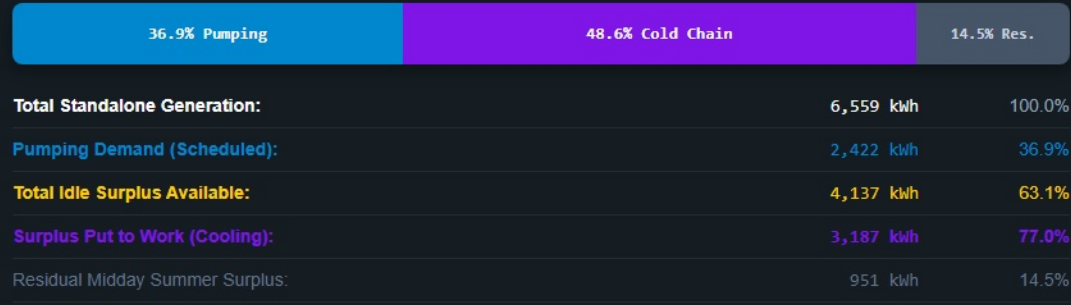
4,137 kWh Surplus Captured at Zero AC Transient Risk

Switching power safely upstream on the 350–600V DC bus eliminates inductive voltage spikes and inverter drive trips during load transfer.

4.8 KWP STANDALONE PV GENERATION PARTITION

Annual Solar Yield: 6,559 kWh / Year

Without AgroStruxure, 63.07% of clean energy is curtailed once pumping halts. AgroStruxure routes 77.04% of this surplus (3,187 kWh/yr) into farm-gate cooling.



FLAWED AC SIDE SWITCHING:
Disconnecting running motor AC leads to inductive back-EMF spikes >1,200V, instant VFD IGBT punch-through & contact arcing.

AGROSTRUXURE UPSTREAM DC:
Controlled VFD deceleration to 0 Hz followed by 5.0s zero-current dead-band ensures contactors switch at exact 0.00 Amps.

Zero Hallucination Mandate: 951 kWh residual summer surplus transparently reported (compressor throttles once PCM fully frozen).

UPSTREAM DC BUS VS RISKY AC SWITCHING 5-Step Zero-Current Transfer Sequence

Deterministic firmware interlock guaranteeing break-before-make switching without contact wear:

- EVENT**

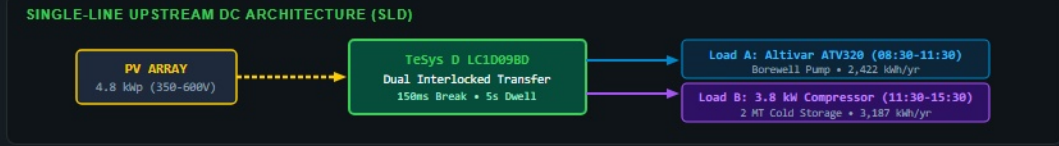
1. Moisture Saturation Cutoff
30cm FDR reaches 45% Field Capacity or daily quota exhausted.
- T = 0.0s**

2. Controlled VFD Deceleration
Modbus Reg 8502 ramps Altivar ATV320 to 0 Hz smoothly over 8.0 seconds.
- T = 8.0s**

3. Zero-Flow Solenoid Closure
1-inch latching valve shuts at zero velocity; zero water hammer shock.
- T = 8–13s**

4. 5.0-Second DC Dead-Band Dwell
Guarantees bus capacitors discharge and DC current reaches 0.00 Amps.
- T = 13.2s**

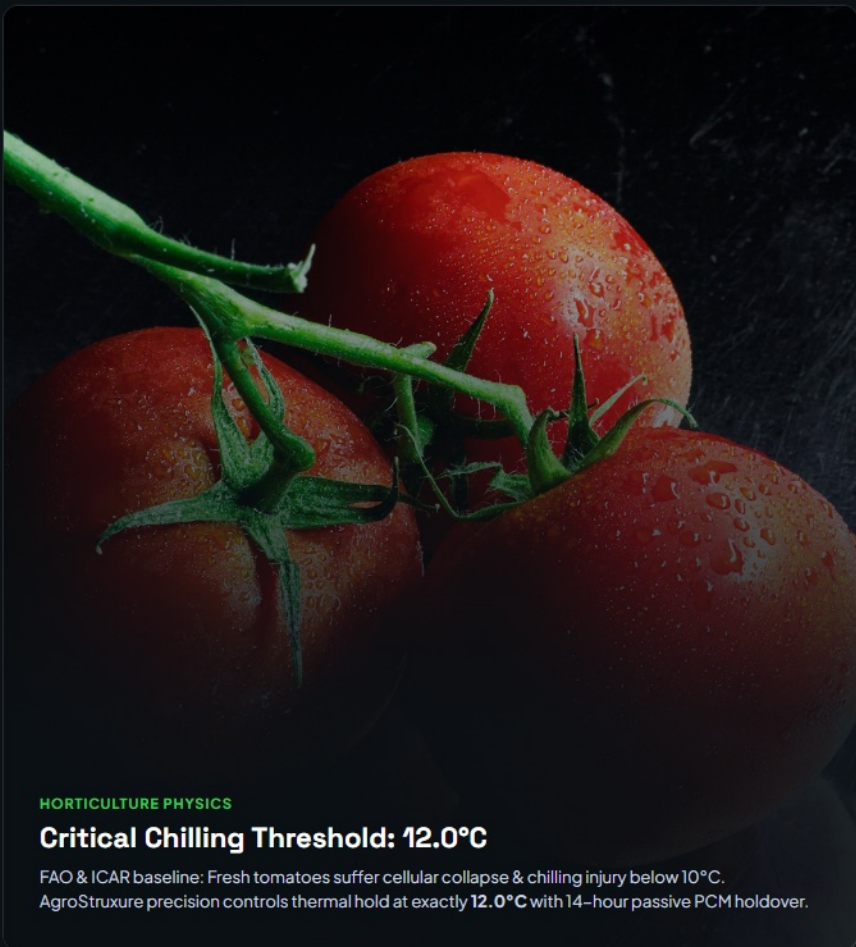
5. TeSys D Contactor Transfer
Dual interlocked contactor energizes 3.8 kW micro-cold room compressor.



Hardware Specification: Schneider Electric TeSys D LC1D09BD (Dual Interlocked) • Type-2 DC SPD Rated 1,000V DC.

Halving Farm-Gate Spoilage from 8.37% to 4.00%

Thermodynamically sized 2 MT farm-gate pre-cooler halts respiration heat within 4 hours, extending tomato shelf life by 4–7 days.



1.31 Tonnes

PRODUCE PRESERVED / HA

Farm-stage spoilage cut by more than half (8.37% down to 4.00%).

₹15,732

DIRECT REVENUE SAVED

1,311 kg grade-A fruit preserved @ ₹12/kg mandi benchmark.

+₹12,000

DISTRESS SALE TIMING

Holding crates for evening mandi pricing (+₹1-₹2/kg on 8 tonnes).

THERMODYNAMIC PULL-DOWN & THERMAL BATTERY

COP = 3.0 • 3.43 kW Cooling Load

Rapid Pull-Down (11:30 AM – 3:30 PM):

Direct solar surplus drives the DC compressor to drop 2.0 MT fresh produce core temperature from 32°C ambient harvest heat to 12.0°C within 4 hours.

Passive PCM Buffer (14-Hour Holdover):

Inorganic salt-hydrate phase change plates freeze at 10°C, maintaining the cold room at 12°C overnight without battery storage or diesel backup.

Net Farmer Gain (Tier A):

₹15,732 (saved fruit) + ₹12,000 (timing) – ₹2,400 (opex)

+₹25,332/yr

Decarbonization:

Replaces diesel cold chain & avoids spoilage loss

2.94 tCO_{2e}

Vernacular AI Grounded in Hardware Truth. Zero Actuation Authority.

Spoken voice advisories in Marathi and Hindi delivered over WhatsApp, grounded 100% in edge telemetry with zero authority to actuate physical switchgear.

DETERMINISTIC AGENTIC ARCHITECTURE

The 4-Stage Cognitive Trust Pipeline

- 1 Edge Telemetry Extraction (Modbus RTU)**
MCU samples FDR sensors (44.8% FC), flow meter (27.4 m³), and Altivar VFD status into circular flash buffer.
- 2 Pydantic Strict Schema Validation**
Telemetry serializes into deterministic JSON. If fields are missing or corrupted, LLM fallback is blocked instantly.

```
{"vwc_30cm": 0.448, "status": "FC_CUTOFF", "pv_kw": 4.21, "load": "PRE_COOLER", "temp_c": 12.0}
```
- 3 Read-Only Deterministic RAG Prompt**
The copilot has read access to verified JSON data and agronomy templates, with strictly ZERO tool execution authority.
- 4 Vernacular Speech Synthesis (TTS)**
Generates high-intelligibility spoken audio in rural Marathi and Hindi dialects delivered via WhatsApp voice notes.

HARD PHYSICAL SAFETY INVARIANT

LLMs do NOT control physical contactors. All switching decisions are executed exclusively by deterministic FreeRTOS C++ state machines. A manual physical switch on the IP67 box allows instant human override.

Krishi Mitra • WhatsApp Spoken Advisory

11:32 AM • Nashik Cluster

"रामभाऊ, तुमच्या टोमॅटोच्या मुळांना पुरेसे पाणी मिळाले आहे (४४.८%). पंप सुरक्षितपणे बंद झाला असून सौर वीज शीतगृहाकडे वळवली आहे. शीतगृहाचे तापमान १२°C असून टोमॅटो थंड होत आहेत."

Translation: "Rambhau, your tomato roots have reached optimal moisture (44.8%). Pumping stopped safely and solar power is now running the cold room at 12°C."



0:14 / 0:26

"धन्यवादा उद्या पाणी कधी सुरू होईल?" (When tomorrow?)

11:34 AM ✓✓

ZERO-LITERACY DESIGN PARADIGM

Smallholder farmers do not use complex dashboards. Krishi Mitra pushes automated 20-second audio voice notes over WhatsApp at key state transitions (Pumping Done → Cold Storage Online), establishing high cognitive trust.

₹50,212 Net Capex per Farm. Payback in 2.0 Years.

Ultra-lean ₹7,540 edge controller BoM paired with a 4-farm cluster pre-cooler sharing model yields rapid capital recovery across two crop harvests.

TIER A: 4-FARM CLUSTER ASSET SHARING MODEL (2 MT)

Capital Expenditure Waterfall

Gross 2 MT Pre-Cooler Capex (Compressor + PCM)	₹400,000
Less: Shared Solar PV Saving (2.6 kWp array avoided)	-₹91,000
Net System Capital Cost	₹309,000
Less: 35% MIDH / AIF Scheme Capital Subsidy	-₹108,150

Net Capex Per Farm (4-Farm Shared Cluster) **₹50,212**

CAPEX PARTITION PER FARM (₹100,000 BASE)



ANNUAL VALUE ACCRETION ENGINE (PER FARM)

Spoilage Saved
+₹15,732

Distress Timing
+₹12,000

Annual Opex
-₹2,400

Tier B Comparison (5 MT FPO Community Hub): **4.2-Year Payback (Leasing)**

CAPITAL RECOVERY DYNAMICS

Cumulative Cash Flow & 2.0-Year Breakeven

Day 0 (Initial Capex Investment):	-₹50,212
End of Season 1 (Month 6):	-₹37,546
End of Year 1 (Month 12):	-₹24,880
End of Season 3 (Month 18):	-₹12,214
Month 24 (Year 2.0) — FULL CAPITAL PAYBACK:	+₹452 (Net Profit)

BREAKEVEN RECOVERY TRAJECTORY



Industrial Edge Controller BoM (1,000-Unit Scale): **₹7,540 / Unit**

ESP32-S3 (₹420) • Isolated MAX485 (₹180) • Dual FDR Probes (₹760) • SHT31 (₹600) • Pulse Flow Meter (₹1,300) • Schneider TeSys D Contactors x2 (₹2,300) • 24V SMPS (₹380) • 4G LTE Module (₹620) • IP67 Enclosure & SPD (₹980).

Standalone Controller Payback: 1.5 crop seasons from pump maintenance & yield protection.

From Software Digital Twin to 250-Farm Commercial Pilot

Structured execution pathway partnering with India's leading horticulture FPOs and Schneider Electric's rural distribution network.



PHASE 1 • CURRENT

Software Twin & Math Validation

0 Physical Units

- **Mathematical Truth:** 365-day energy, water, and thermal balance derived in `impact_model.py`.
- **Modbus State Machine:** Emulated Altivar ATV320 Command (8501) and Speed (8502) registers.
- **Vernacular NLP:** Pydantic prompt guard verified for Marathi & Hindi voice note synthesis.
- **Zero-Cloud Guarantee:** Deterministic C++ state machine runs without cellular dependency.

Exit Gate: Zero discrepancies in mathematical mass and thermal balances.

PHASE 2 • Q1-Q2 2027

Hardware-in-the-Loop Testbench

1 Testbench

- **Physical Bench:** ESP32-S3 paired with Schneider Altivar ATV320 solar VFD & motor dynamometer.
- **Switchgear Safety:** 5.0s DC dead-band dwell verified under 500V DC break conditions.
- **Surge Resilience:** Type-2 DC SPD endurance testing against 1,000V DC lightning transients.
- **Inverter Protection:** Validated zero IGBT punch-through across 500 repeated transfer cycles.

Exit Gate: 500 faultless DC transfer cycles under full electrical load without drive trip.

PHASE 3 • Q3-Q4 2027

20-Farm FPO Pilot (Nashik)

20 Farms (5 Clusters)

- **Field Partnership:** Deployed across 5 clusters (4 farms each) with Sahyadri Farmers Producer Co.
- **Real Tomato Harvest:** Continuous FDR root monitoring & pre-cooling connected to reefer transport.
- **Farmer Trust Loop:** Daily spoken WhatsApp notes achieving 98%+ farmer comprehension.
- **Agronomic Cutoff:** Validated collar rot prevention across two full Solanaceae harvest cycles.

Exit Gate: Measured >35% water conservation & 95%+ pre-cooler uptime over 2 seasons.

PHASE 4 • 2028

250-Farm Commercial Expansion

250 Farms (5 FPOs)

- **OEM Retrofit Skids:** Factory-assembled DIN rail kits reducing solar pump EPC install time to 2 hours.
- **"Urja Mitra" Network:** Local ITI diploma youth trained for sensor calibration & field servicing.
- **Carbon Revenue:** Aggregating 2.94 t CO₂e/ha/yr for verified voluntary carbon credit issuance.
- **Channel Synergy:** Pre-integrated as catalog option with Schneider Electric Rural EPC network.

Exit Gate: Commercial breakeven with 5 FPO enterprise contracts in Maharashtra & Rajasthan.

One Solar Pump. Four System Outcomes.

AgroStruxure creates a high-margin digital IoT layer for Schneider Altivar Solar drives and TeSys switchgear, turning agricultural clean-tech from standalone pumps into connected community microgrids.

41.2%

WATER CONSERVED

7,000 m³/ha/yr aquifer water saved. Root-zone hypoxia eliminated, stabilizing critical groundwater tables.

4,137 kWh

CLEAN ENERGY CAPTURED

Surplus solar redirected upstream on DC bus into 2 MT cold storage, avoiding dedicated solar capex.

1.31 Tonnes

PRODUCE PRESERVED

Farm-stage spoilage halved from 8.37% to 4.00%. First-mile pre-cooling extends shelf life by 4–7 days.

2.0 Years

CAPITAL PAYBACK

₹50,212 net capex recovered in 2 crop harvests. Generates + ₹25,332/farm/yr in ongoing net profits.

Sanskar Tiwari

EMBEDDED FIRMWARE & LEAD

FreeRTOS deterministic state machine, Modbus RTU, CIA402 drive control.

Shambhavi Patil

GEOSPATIAL INTELLIGENCE

Sentinel-2 MSI 10m NDVI/NDWI, dynamic FAO-56 dual Kc curves, aquifer balance.

Kanishka Salgude

ELECTRICAL SYSTEMS & VFD

Schneider Altivar ATV320 VFD configuration, TeSys D interlocking, DC power routing.

Chaitanya Ranade

PRODUCT STRATEGY & AGRONOMY

Sahyadri FPO cluster economics, MIDH/AIF subsidies, vernacular WhatsApp bot UX.

Schneider Electric Strategic Synergies: Altivar Solar Drives OEM Pre-Installation • EcoStruxure Microgrid Edge Integration • SE Ventures Cleantech Scale

Net Decarbonization: 2.94 t CO₂e / ha / year